

Dunder Mifflin Employee Attendance Analytics

Portfolio Project | Data Analyst / MIS Analyst

Executive Summary

This project converts raw employee In/Out attendance events into an HR analytics dataset and management reporting layer. The analysis focuses on attendance completeness, work hours, break time, late arrivals, employee-level patterns, department-level totals, and daily trends.

Business Problem

Raw attendance logs contain individual access events rather than the metrics an HR or operations manager needs. The objective is to transform those events into understandable KPIs and exception indicators.

Dataset & Scope

- 14 employees across 6 departments.
- 528 raw attendance events across 5 working days.
- Date range: 04 Nov 2019 to 08 Nov 2019.
- Analysis grain: employee-day, with employee, department and daily rollups.

Technology Stack

Excel for reporting and validation; SQL for aggregation; Python/Pandas for data transformation and analysis; Power BI as the recommended visualization layer.

Data Preparation & Methodology

1. Sorted raw events by employee and timestamp.
2. Grouped events by employee and date.
3. Used the first In event and last Out event to define the attendance span.
4. Summed Out-event durations to calculate recorded work hours.
5. Calculated break/non-work time as attendance span minus work hours.
6. Created a late-arrival flag for first entry after 09:00 AM.



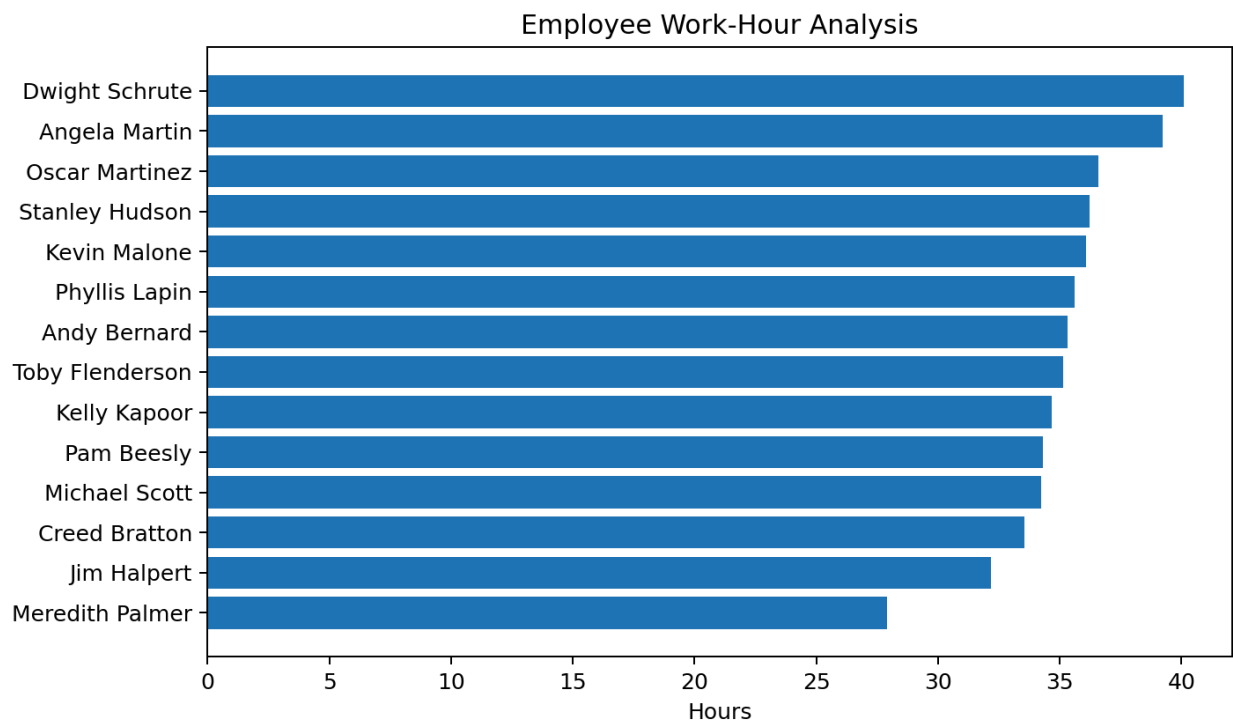
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7. Checked missing values and employee-day completeness before analysis.

Key Results

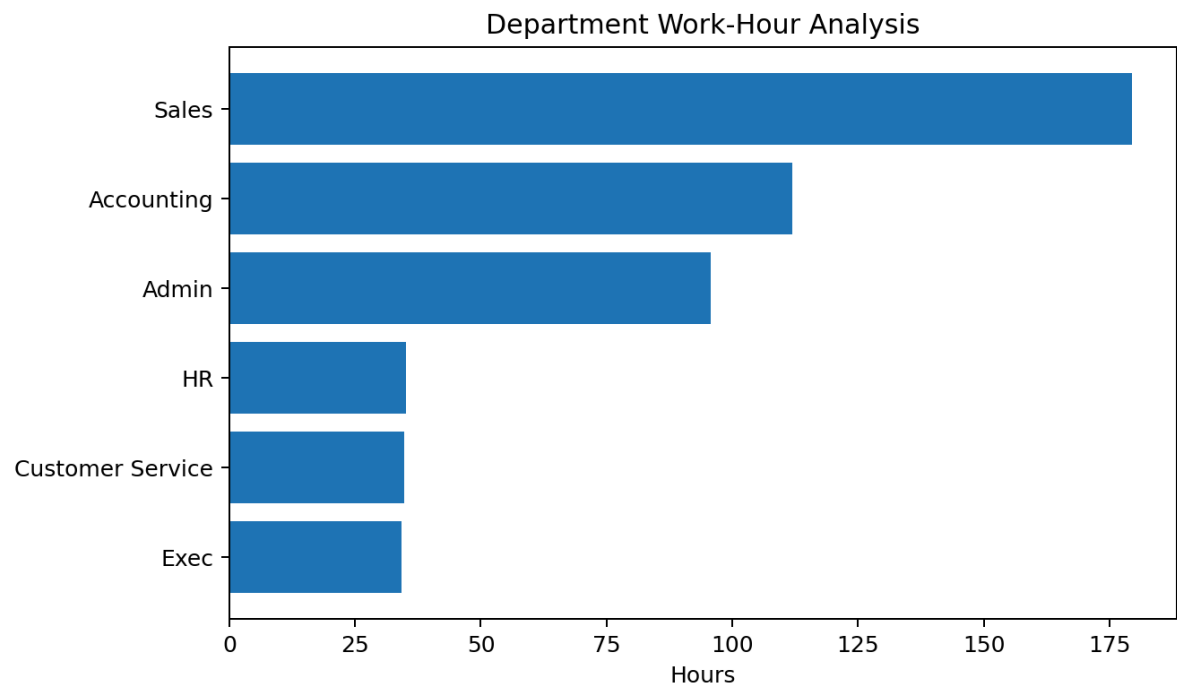
- Total recorded work hours: 491.15 hours.
- Average work hours per employee-day: 7.02 hours.
- Average break/non-work span per employee-day: 1.75 hours.
- Late arrivals after 09:00 AM in the five-day sample: 0.
- All 14 employees have records for all five working days in this dataset.

Employee Analysis



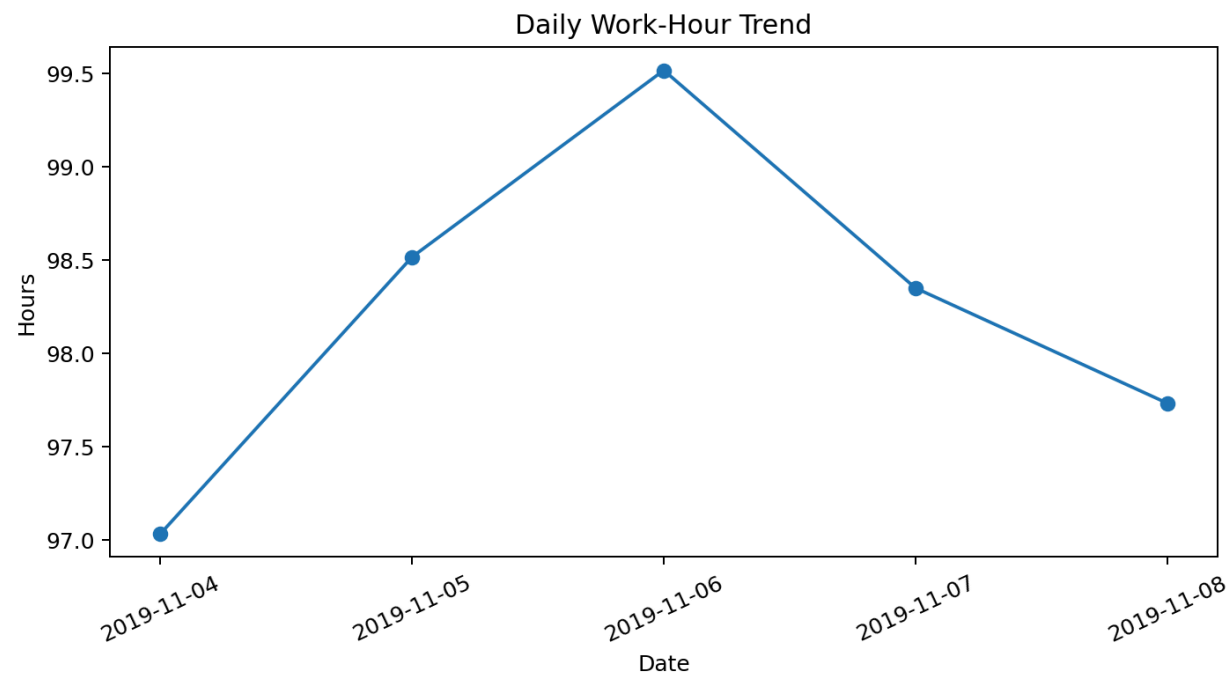
The employee-level summary allows HR/operations teams to compare total work hours, average daily hours, break time and late-arrival counts without manually reading event logs.

Department Analysis



Department aggregation provides a staffing/work-hour view that can be used for operational reporting and workforce planning.

Daily Trend



The daily trend shows how total recorded work hours changed during the five-day observation window.

Data Quality

- Source rows: 528 – Attendance event rows in source
- Employees: 14 – Distinct employees
- Departments: 6 – Distinct departments
- Date range: 04-Nov-2019 to 08-Nov-2019 – Coverage
- Blank Prev Time: 13 – Expected for first event per employee/day
- Missing Employee: 0 – Should be 0
- Missing Department: 0 – Should be 0
- Missing DateTime: 0 – Should be 0
- Employee-days: 70 – Expected employee/date combinations
- Employee-days with both In and Out: 70 – Data completeness

Business Recommendations

- Build a Power BI executive page with headcount, total work hours, average hours, break time and late-arrival KPIs.
- Add department and employee slicers for drill-down analysis.
- Create an exception table for late arrivals, unusually low daily hours, missing In/Out pairs and anomalous durations.
- For production use, define attendance thresholds with HR policy rather than treating this sample's thresholds as policy.
- Expand the data window to several months before using the analysis for trend or workforce-performance decisions.

Interview Answer – 60 Seconds

"I worked on an employee attendance analytics project using event-level In/Out data. The raw data was difficult to use directly, so I cleaned and sorted the records and transformed them into an employee-day dataset. I calculated first-in, last-out, work hours, break time and late-arrival flags, then aggregated the data by employee, department and date. I used Python/Pandas for transformation, SQL for analytical queries, and Excel/Power BI for reporting. The final dashboard is designed to help HR and operations teams monitor attendance and work-hour patterns and quickly identify exceptions. I also



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performed data-quality checks before presenting the KPIs.”

Portfolio Disclaimer

This is a portfolio/learning project based on a five-day sample. The analysis demonstrates the workflow and technical skills; it should not be interpreted as a long-term employee performance assessment.



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